

Geology of the Payer-Weyprecht Mountains, Central Dronning Maud Land, East Antarctica

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Abstract

Orthogneiss with tabular to lensoid pyroxene-amphibolite enclaves, is exposed in the ice-free areas of Payer and Weyprecht Mountains. Orthogneiss was emplaced late syntectonic to D2 deformation under amphibolite facies conditions which imparted a homogeneous foliation. The field relations and geochemical parameters suggest an I-type affinity for the granitic orthogneiss, while the pyroxene-amphibolite enclaves are metamorphosed products of basic igneous rocks.

Introduction

Payer-Weyprecht Mountains and associated nunataks are exposed about 80 km southeast of Gruber massif (Wohlthat mountains) in the Central Dronning Maud Land, East Antarctica (Fig 1). The outcrop free area between the two mountain ranges is occupied by thick continental ice sheet. Geological mapping of these distant mountains covering an area bounded by latitudes 71°49'36" & 72°05'48"S and longitudes 13°09'34" & 14°49'27"E, was undertaken during the eleventh Indian expedition to Antarctica.

Geology

Regional geological setting :

There is hardly any geological information available from Payer-Weyprecht region, however, the adjoining Wohlthat mountains are fairly well studied (Ravich and Kamenev, 1975, Kaul et al., 1988, Ravindra et al., 1989, Joshi and Bejarniya, 1990). The geological account of Schirmacher hill range which lies further about 90 km north of Wohlthat mountains has also been attended to in detail by Kaul et al. (1985), Singh (1986) and Sengupta

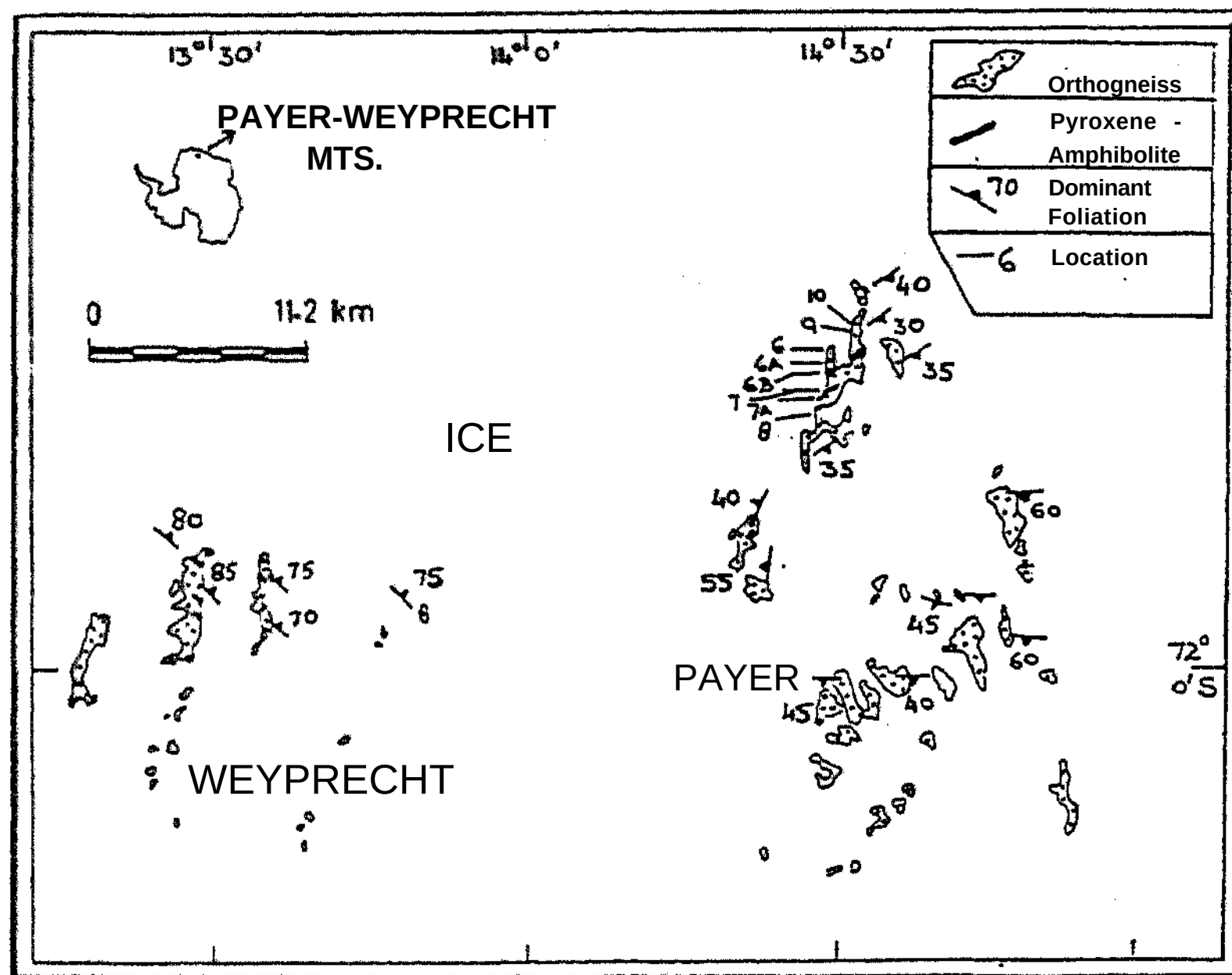


Fig 1: Geological Map of Payer-Weyprecht Mountains, East Antarctica.

(1988,1993). All these studies have indicated the presence of a paragneissic complex with interlayered orthogneiss and mafic granulite. This gneissic complex has undergone multiphase tectonothermal events (Joshi et al., 1991a, Sengupta op.cit.) and is intruded by anorthosite, monzogabbro/monzonite and many plutons of post-tectonic alkali granite. The alkali granite has been inferred to have A-type affinity and probably manifest a Pan-African thermal event (Joshi et al., 1991b, Joshi and Pant, 1994). Paucity of geochronological data from this region does not enable assignment of exact ages, but on the basis of extrapolation of data from adjoining regions having similar lithology and tectonothermal history, a mid-Proterozoic age has been suggested (James and Tingey, 1983).

In the Payer-Weyprecht mountains, a quartzofeldspathic biotite-hornblende gneiss is a dominant rock type containing enclaves of pyroxene-amphibolite, which is occasionally garnetiferous.

A coarse-grained undeformed granitoid was mapped in nunataks southeast of Weyprecht. One doleritic sill and basaltic dyke intruding the gneiss was observed apart from numerous pegmatite/granite veins. Field relations, petrography and petrochemical studies of granitic orthogneiss and amphibolite enclaves are discussed.

Field characteristics and relations:

Gneiss is homogeneously foliated (Fig 2), the foliation being defined by the preferred shape orientation of biotite and hornblende. Composition layering is variably developed. In contrast the mafic enclaves which are layered and foliated, occur as discontinuous tabular bodies and boudins of variable dimensions with sharp but occasionally sheared contacts. Mutual angularity between foliation in gneiss and those within enclaves is generally absent. Gneiss is free from any intercalation with metasedimentary rock throughout its exposure. Foliated apophyses and veins invading pyroxene amphibolite enclaves at the contact indicate an intrusive nature of the former (Figs 3,4b & 4d). The intrusive nature is further strengthened by absence of an early isoclinal phase of folding in gneiss which is present in the amphibolite enclaves. The mafic enclaves show well preserved layering and isoclinal rootless folds (Fig 4a) indicative of an earlier tectonothermal event. Four sets of pegmatite/granite veins have been profusely injected into these rocks under brittle ductile conditions (Fig 4c).

Petrography:

Orthogneiss: It is a medium-grained, leuco- to mesocratic and uniformly foliated rock. The mineral assemblage recorded is — Quartz-orthoclase & microcline perthites-oligoclase-hornblende- biotite-zircon-sphene-garnet.

The rock shows granoblastic textures with minor differentiation. Strong shape preferred alignment of biotite, hornblende and elongated quartz define the foliation (Fig 5). Quartz grains show lobate to serrate boundaries attesting grain boundary migration and diffusion during amphibolite facies metamorphism. Quartz shows sub-grain development (Fig 6), polygonization and deformation bands. Microcline grains exhibit deformed grid twins while bent polysynthetic twinning and undulose extinction are noted in plagioclase grains. Development of myrmekite embaying into microcline perthite is also seen (Fig 7). These textural features indicate imprints of post-crystalline deformation. Further, euhedral zircon grains are recorded throughout the granitic body (Fig 8a-d).

Pyroxene-amphibolite:

They are well foliated and occur as enclaves within orthogneiss. The quartzofeldspathic materials are present in some enclaves. The mineral assemblage observed is —

Hornblende-biotite-plagioclase+perthite-clinopyroxene+orthopyroxene quartz-sphene-ilmenite.

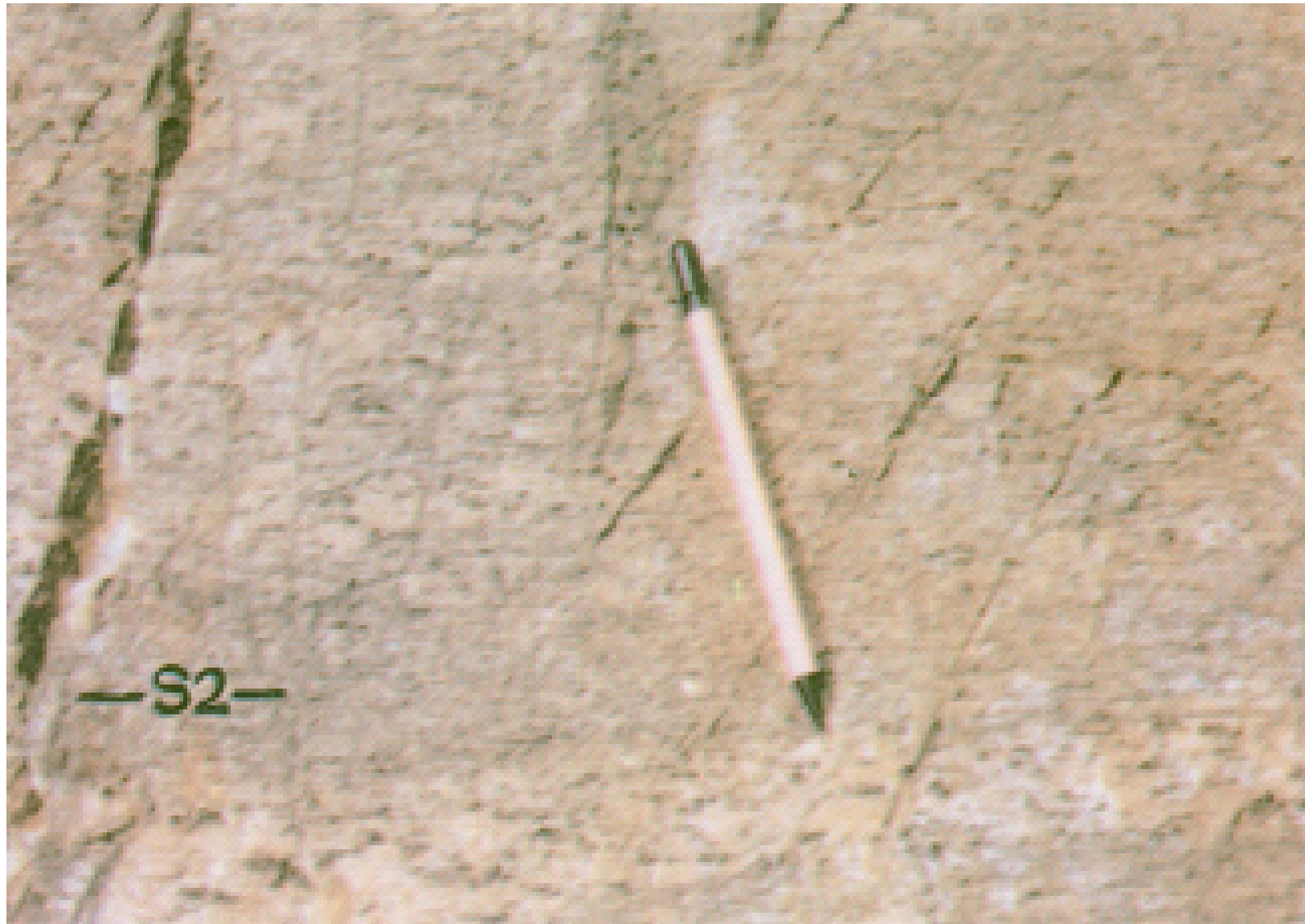


Fig 2: Uniformly foliated granite, foliation defined by biotite and hornblende grains/grain aggregates.



Fig 3 : Layered mafic enclave in orthogneiss. Intrusive contact relation shown by apophyse of orthogneiss invading the pyroxene amphibolite enclave.

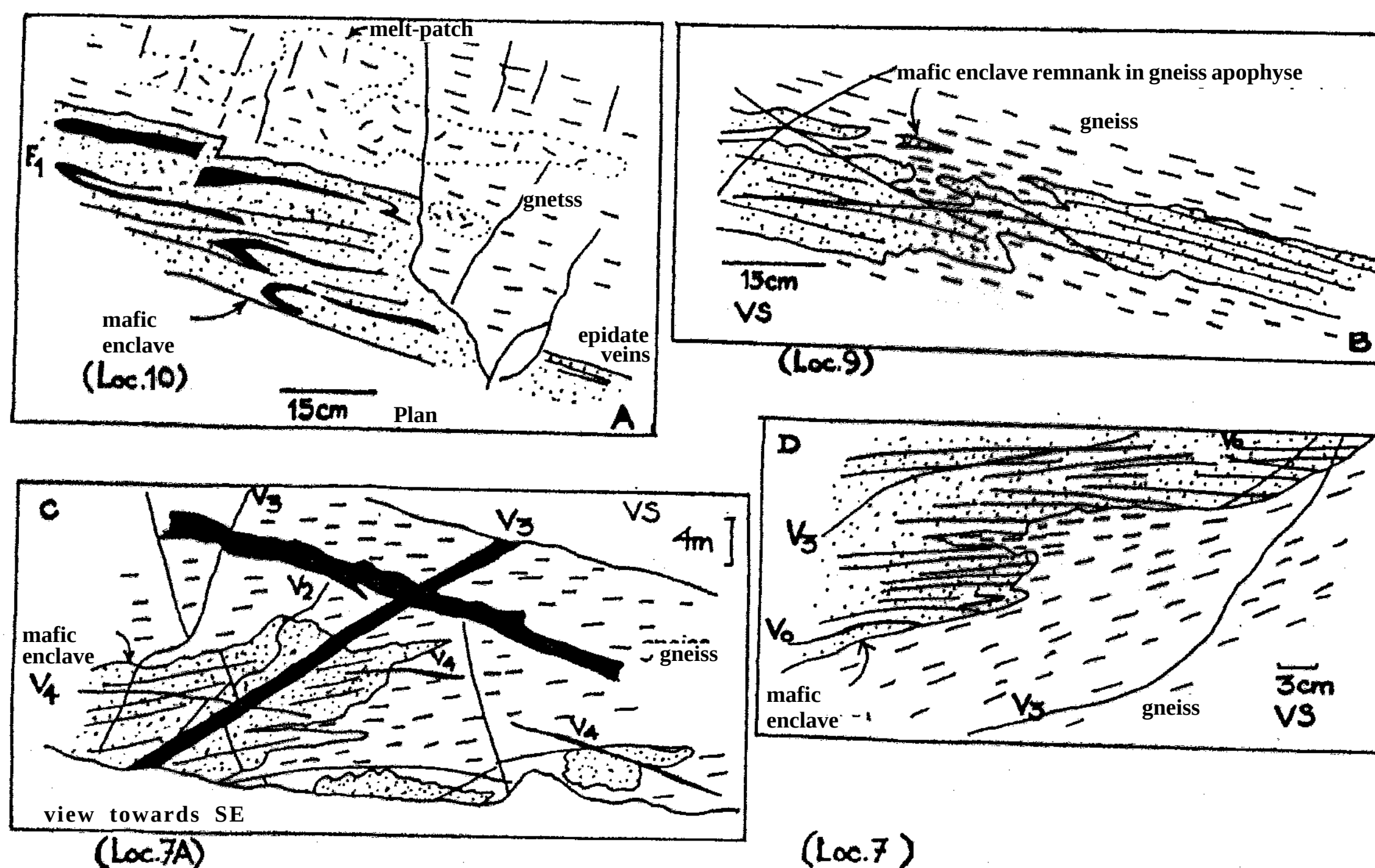


Fig 4 : a) Contact between orthogneiss, showing quartzofeldspathic melt patches, and mafic enclaves showing isoclinal folds on quartzofeldspathic layering. Stippled pattern represents mafic enclave; b) Contact of mafic enclaves with orthogneiss showing metagranite (orthogneiss) invading mafic enclave; c) Field view of mafic enclave in orthogneiss: note intrusive contacts and V₂, V₃, and V₄ generation of pegmatite/aplite veins; d) View of intrusive nature of orthogneiss (metagranite) invading mafic enclave : note truncated quartzofeldspathic layering (V₀) in enclave.

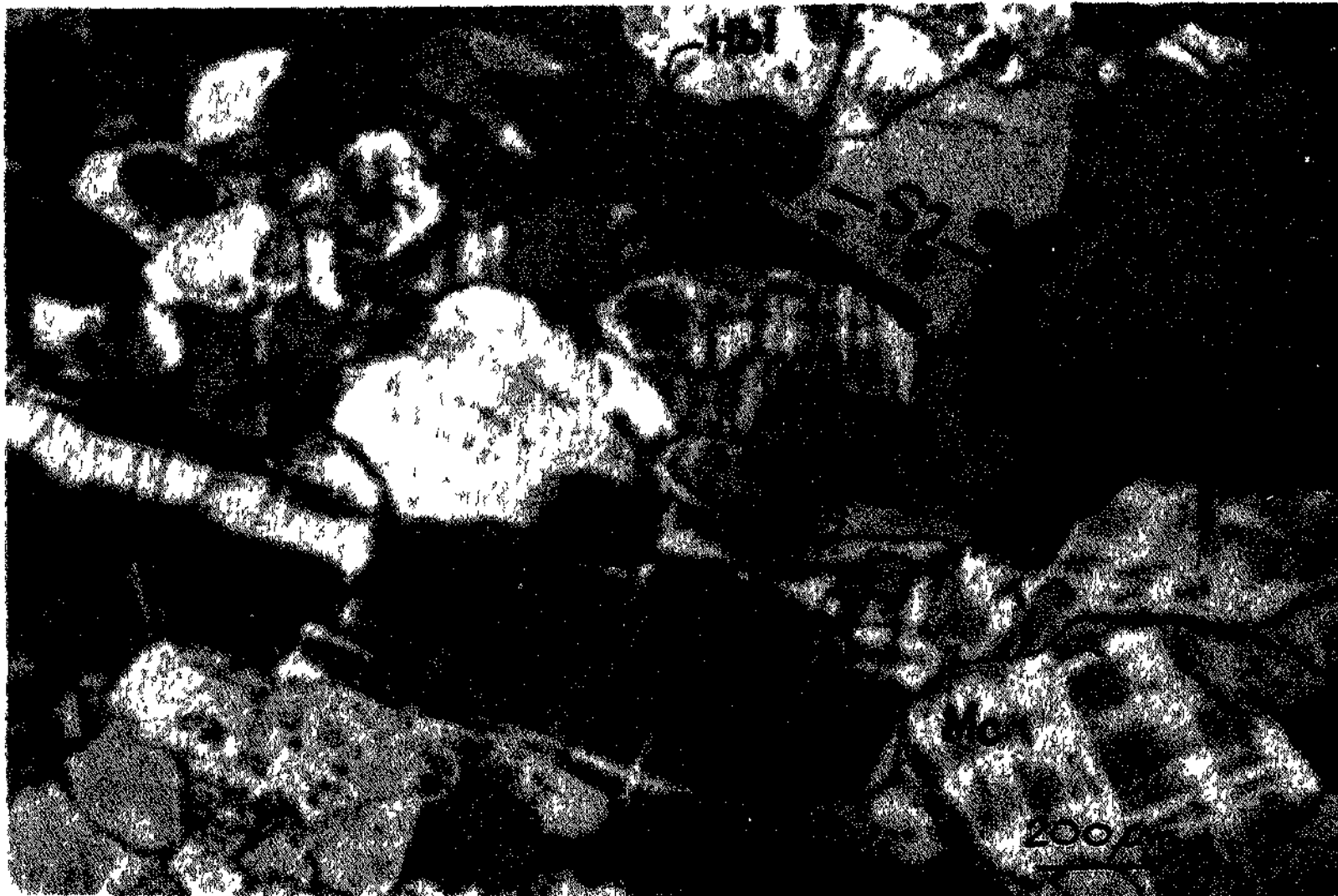


Fig 5 : Orthogneiss showing preferred alignment of biotite-hornblende and elongated perthite defining foliation.



Fig 6 : Orthogneiss showing sub-grain development in quartz

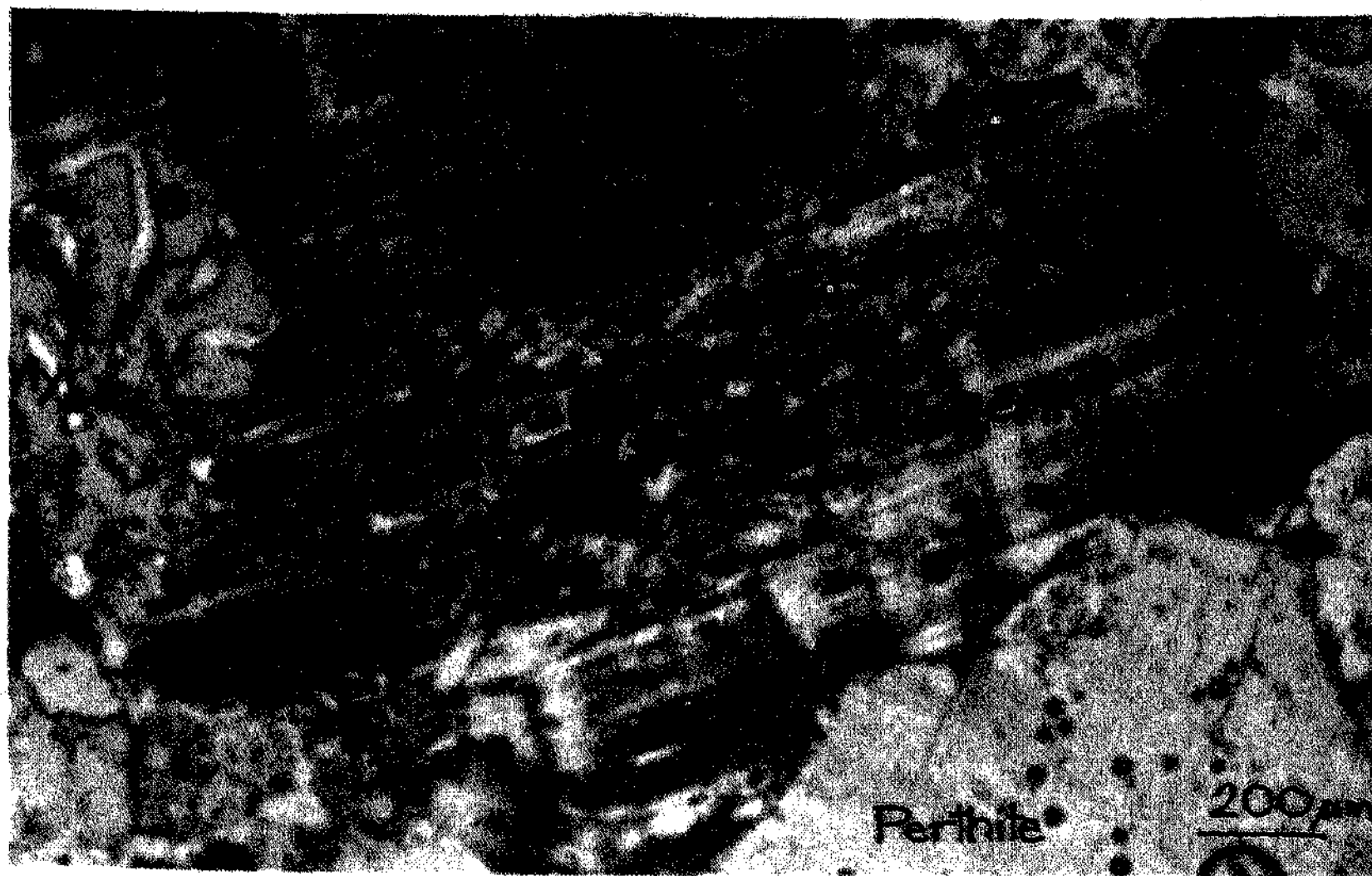


Fig7: Myrmekite embaying into microcline microperthite in orthogneiss.

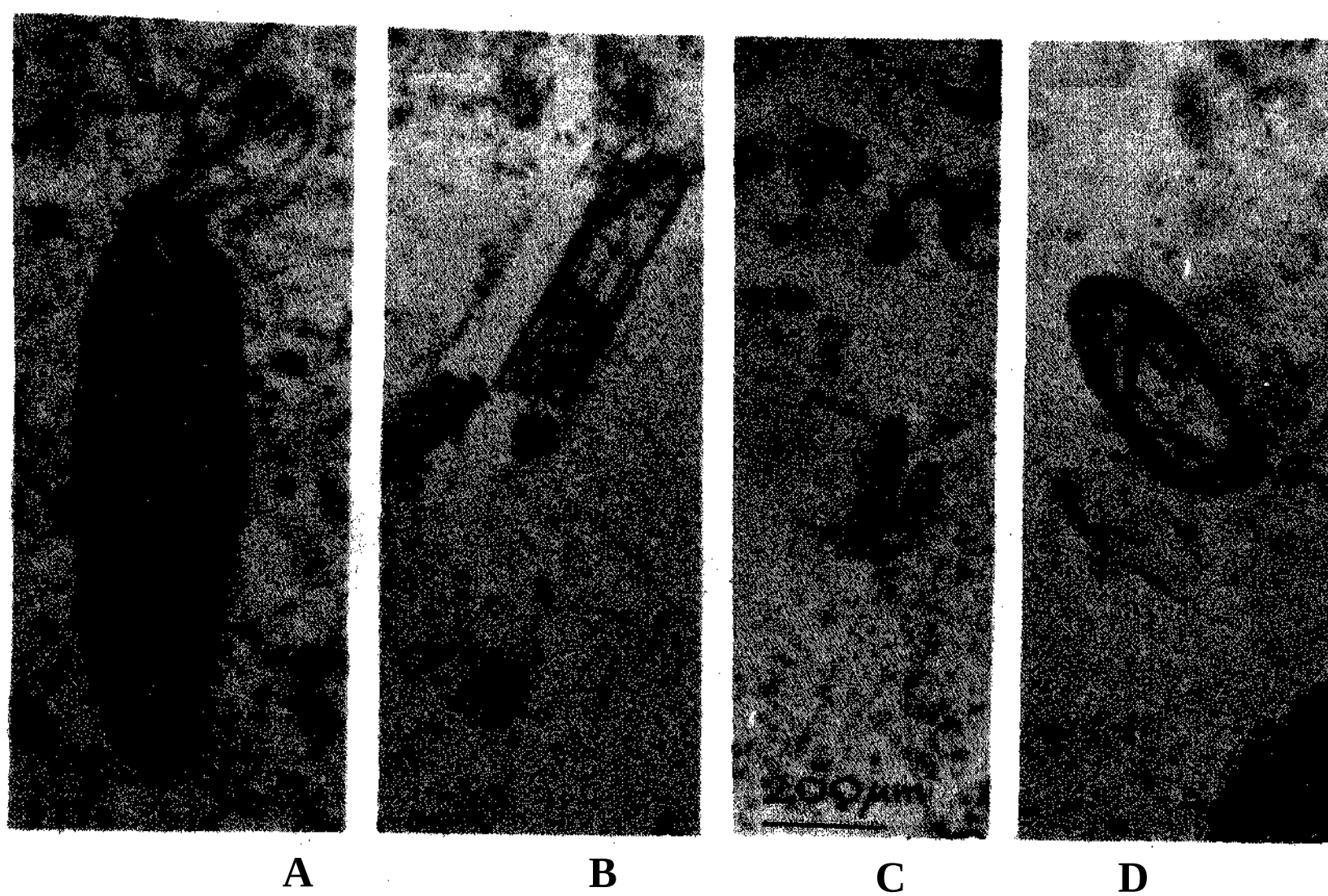


Fig 8: (a-d) Euhedral zircon grains in granitic orthogneiss.

These amphibolites wherever well assimilated by the host gneiss appear like grey coloured mafic gneiss bands. Quartz-plagioclase layers trace rootless isoclinal folds in these amphibolites. Quartz-microcline-plagioclase granitic veins indicate later injection during the granitic emplacement, cross cutting the earlier quartz-plagioclase layers. In thin section the foliation is defined by aligned biotite and hornblende and elongate quartzofeldspathic aggregates. Clinopyroxene and orthopyroxene grains are seen to be retrogressing to fibrous amphibole and Mg- rich chlorites. Sphene coronas around ilmenite (Fig 9) also indicate retrogression from granulite to amphibolite facies.

Coarse-grained granitoid intrusive:

They occur in nunataks southeast of Weyprecht and comprise the following mineral assemblage — Quartz-perthite-oligoclase-biotite-hornblende-orthopyroxene± clinopyroxene- zircon-opaques

This pinkish coloured chamockitic rock shows typical hypidiomorphic texture (Fig 10) with little deformation at grain boundaries. Patch type and string type orthoclase — perthites dominate over oligoclase grains in this rock.

Mafic intrusives:

These are represented by minor dolerite sills and basaltic dykes and show the following mineral assemblage — Plagioclase-hornblende-biotite-quartz-orthoclase±epidote-sphene-zircon-opaques

Texture seen is relict intergranular with imprints of low grade (greenschist facies) metamorphism.

Pegmatite/granite veins:

Four sets, V1 to V4 veins were observed (Fig 4c), with V3 and V4 veins post-dating D4 deformation. V1 and V2 vein sets have been timed as pre-D3 and D4, respectively. Their mineral composition is similar to that of host granitic orthogneiss into which they intrude. The mineral assemblage is —

Quartz-orthoclase perthite-microcline-oligoclase-hornblende- biotite-zircon-magnetite-ilmenite

The aplitic and pegmatitic veins in turn cross-cut the granite intrusives and are thought to be dilatational veins.

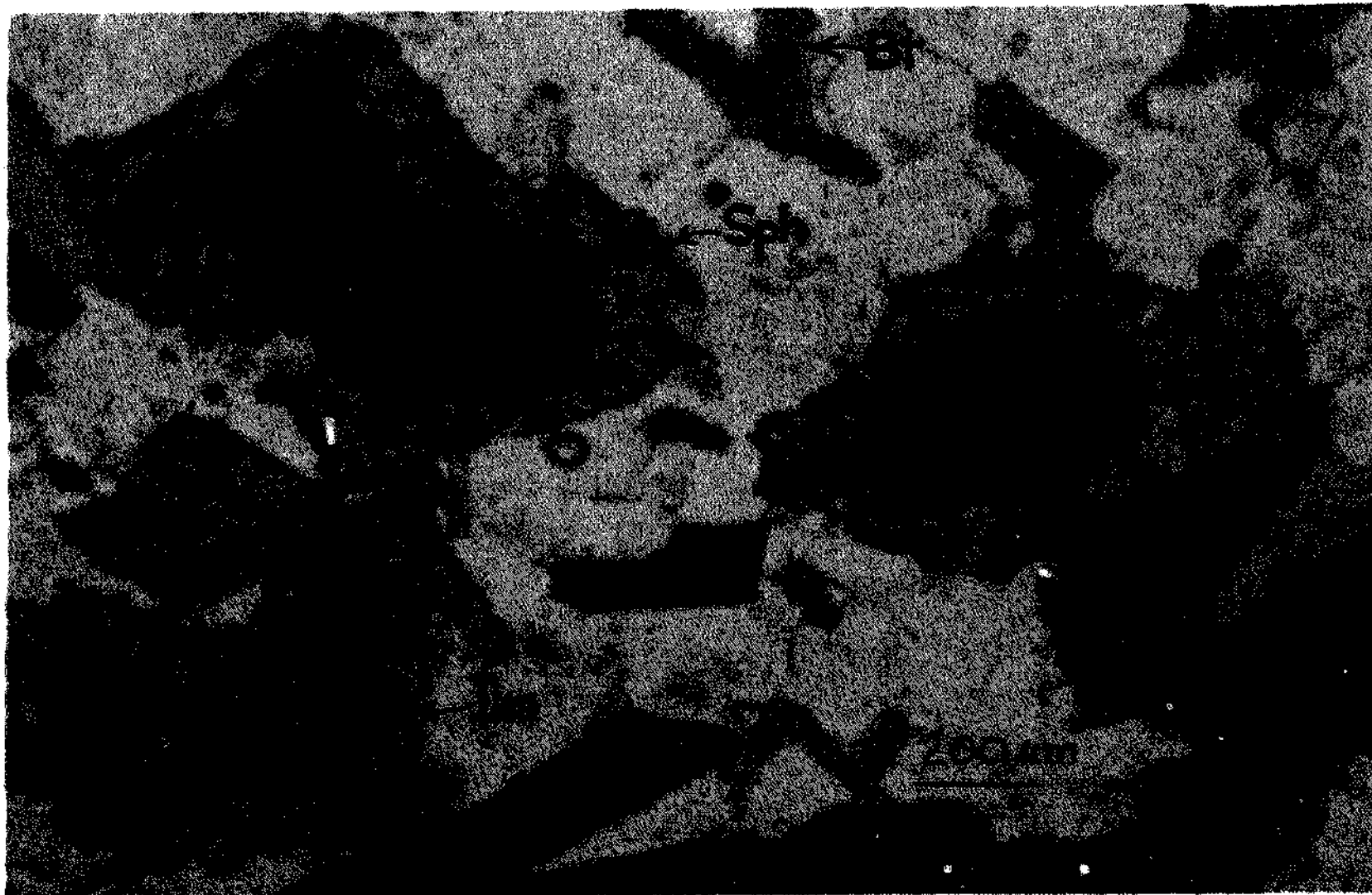


Fig 9: Sphene coronas around ilmenite grains in pyroxene- amphibolite.

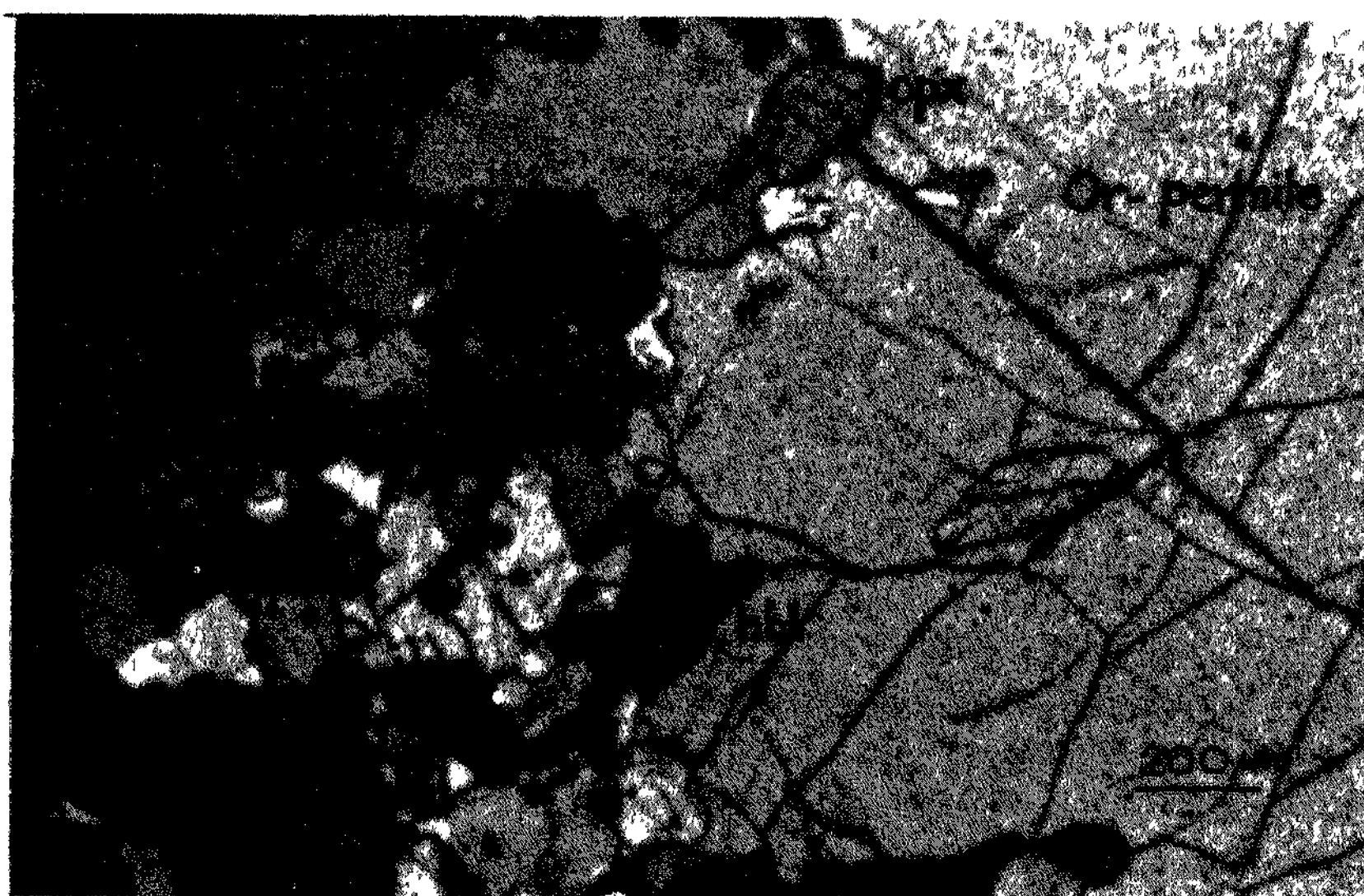


Fig 10: Orthopyroxene bearing intrusive charnockite of Weyprecht.

Structure and metamorphism

The foliation in Payer mountains trends ENE with a 35° to 60° dip towards SSE whereas in Weyprecht range it strikes in a NNW direction with steep dips towards ENE.

Field observations on mesoscopic folds and their mutual relations suggest a polydeformational history. The earliest deformation event D1 produced F1 folds of isoclinal reclined geometry traced by thin quartzofeldspathic layers (Fig 4a). They are preserved only within the mafic amphibolite enclaves. Second deformation phase D2 produced F2 folds which are coaxial and coplanar with F1 (Figs 11a & 11b). The F1 and F2 folds are confined only to amphibolite enclaves and have their axial planes trending in a NE-SW direction, whereas the surrounding orthogneiss has developed a pervasive foliation which is nearly parallel to axial plane of F2 folds. This field relation clearly suggests a broadly syntectonic emplacement of orthogneiss with respect to D2 event (at amphibolite facies metamorphic conditions).

D3 and D4 events have affected both the amphibolite and the orthogneiss (Figs 11c and 11d). Imprints of these events are well manifested by the folding

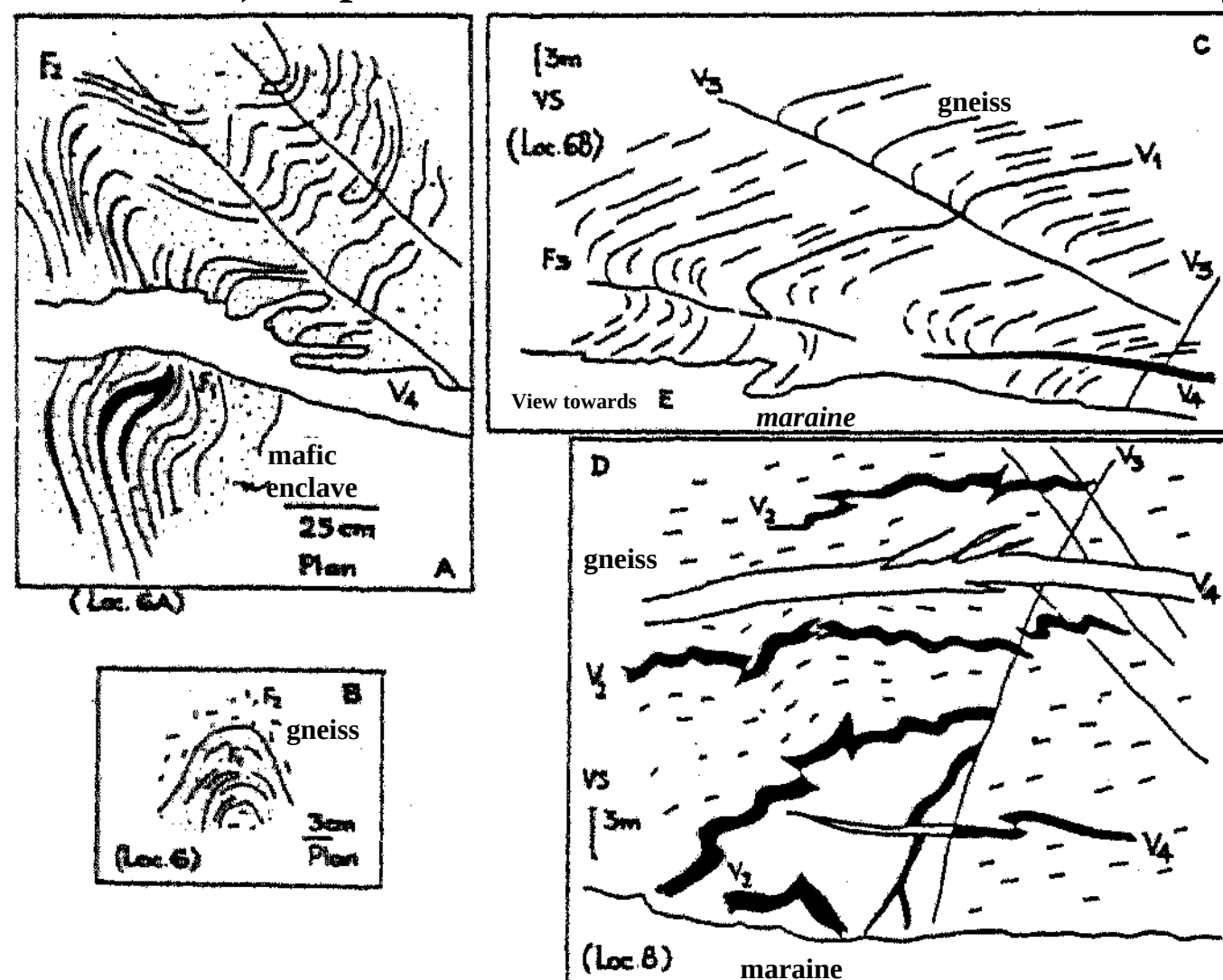


fig 11: a) F2 and minor preserved F1 folds on quartzofeldspathic layering in mafic enclaves; b) Refolding of F1 by F2 in mafic enclave; c) Field view of F3 horizontal plunging shear related fold in orthogneiss: note V1 pegmatite veins folded by F3 and cross cutting V3 and V4 pegmatite/aplite veins. V1 pre-dates F3 event but post-dates F2 event; d) F4 upright folds affecting V2 aplite/pegmatite veins being cross cut by V3 and V4 veins.

of pegmatite/aplite veins. D3 event has given rise to horizontal plunging to plunging inclined F3 shear related folds. Open upright F4 folds trending NW-SE have developed as a result of layer parallel compression during D4.

Metamorphism:

Metamorphic grade has been inferred from the mineral assemblage of mafic amphibolite enclaves. These show following assemblage indicating amphibolite facies metamorphism :

Hornblende-plagioclase-biotite-clinopyroxene±orthopyroxene-sphene-ilmenite-magnetite

Presence of relict orthopyroxene (Fig 12), fibrous amphibole pseudomorphing pyroxene and sphene coronas replacing ilmenite (Fig 9) indicate an earlier granulite facies mineralogy overprinted by a later lower grade amphibolite facies metamorphism.

Further, the biotite-hornblende foliation (S2) (Fig 5) in the orthogneiss represents a high T steady state foliation. This foliation is parallel to the amphibolite facies hornblende-plagioclase-biotite foliation in the amphibolite enclaves, indicating development of foliation at amphibolite facies.

Late greenschist facies imprint (post-dating D4) show garnet retrogressing to biotite, chlorite and quartz aggregates (Fig 13) and replacement of basaltic assemblage by a quartz-chlorite-calcite-opaque oxide assemblage.

Geochemistry

The whole rock powdered samples of orthogneiss and amphibolite were analysed for major oxides by XRF using PW 1410 (Philips) in the AMSE Laboratory, Geological Survey of India, Bangalore. The analyses obtained along with normative mineralogy, are presented in Tables I and II.

Orthogneiss

The SiO₂ content varies from 69 to 75.05%, Na₂O is lower and K₂O is high, while remaining major oxide contents are comparable with average composition of granite and granodiorite. Appearance of hypersthene and quartz in absence of diopside in the norms indicate silica oversaturated nature. Diopside appears in norms where there is significantly high presence of Fe₂O₃ and CaO. Smooth variation in Al₂O₃, TiO₂, and weak CaO and Na₂O correlation with respect to silica seen in the Harker diagram may reflect melting of a homogeneous source (Fig 14).



Fig 12: Relict orthopyroxene within hornblende-plagioclase assemblage in pyroxene amphibolite.

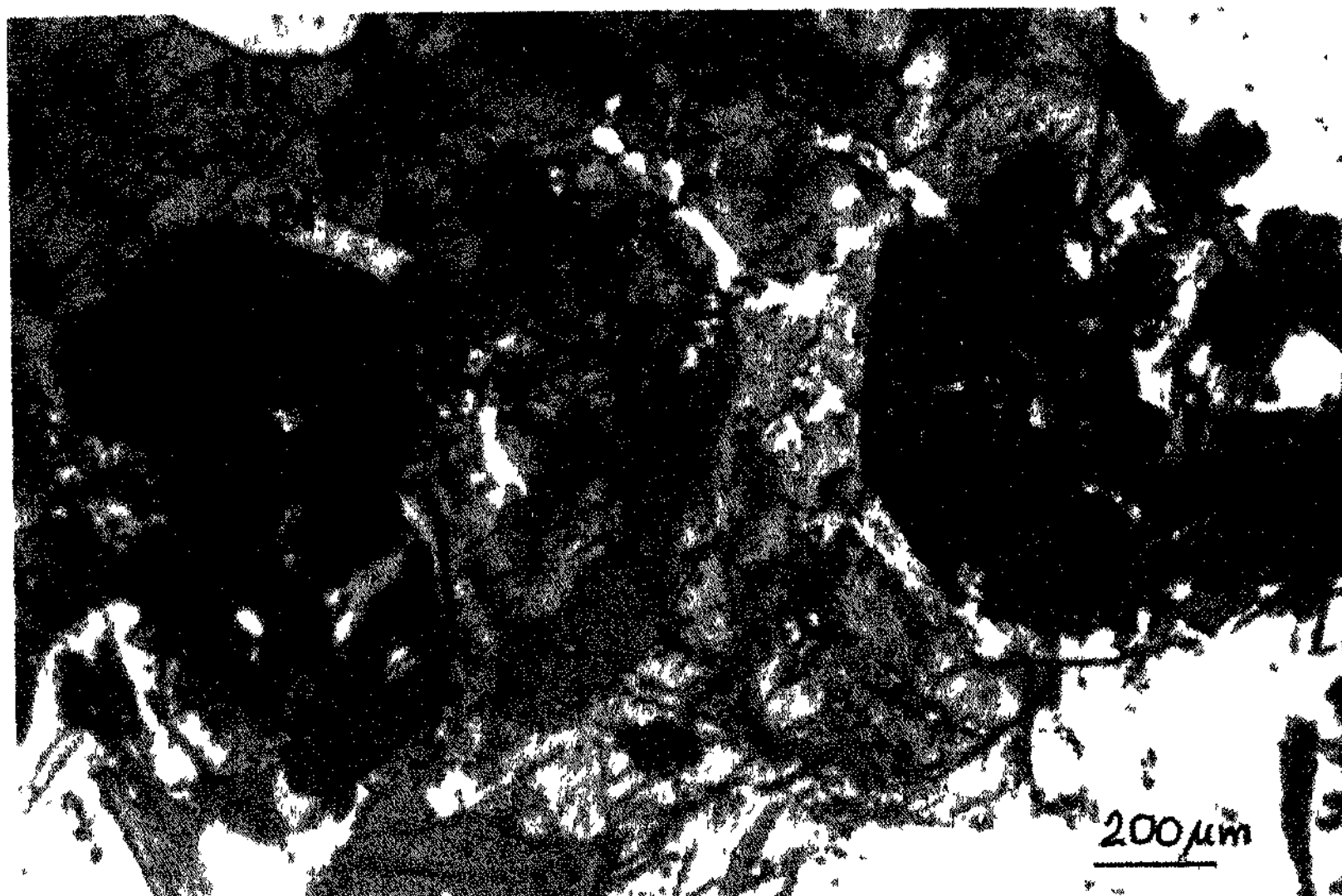


Fig 13 : Garnet retrogressing to biotite, chlorite and quartz aggregates in granitic orthogneiss.

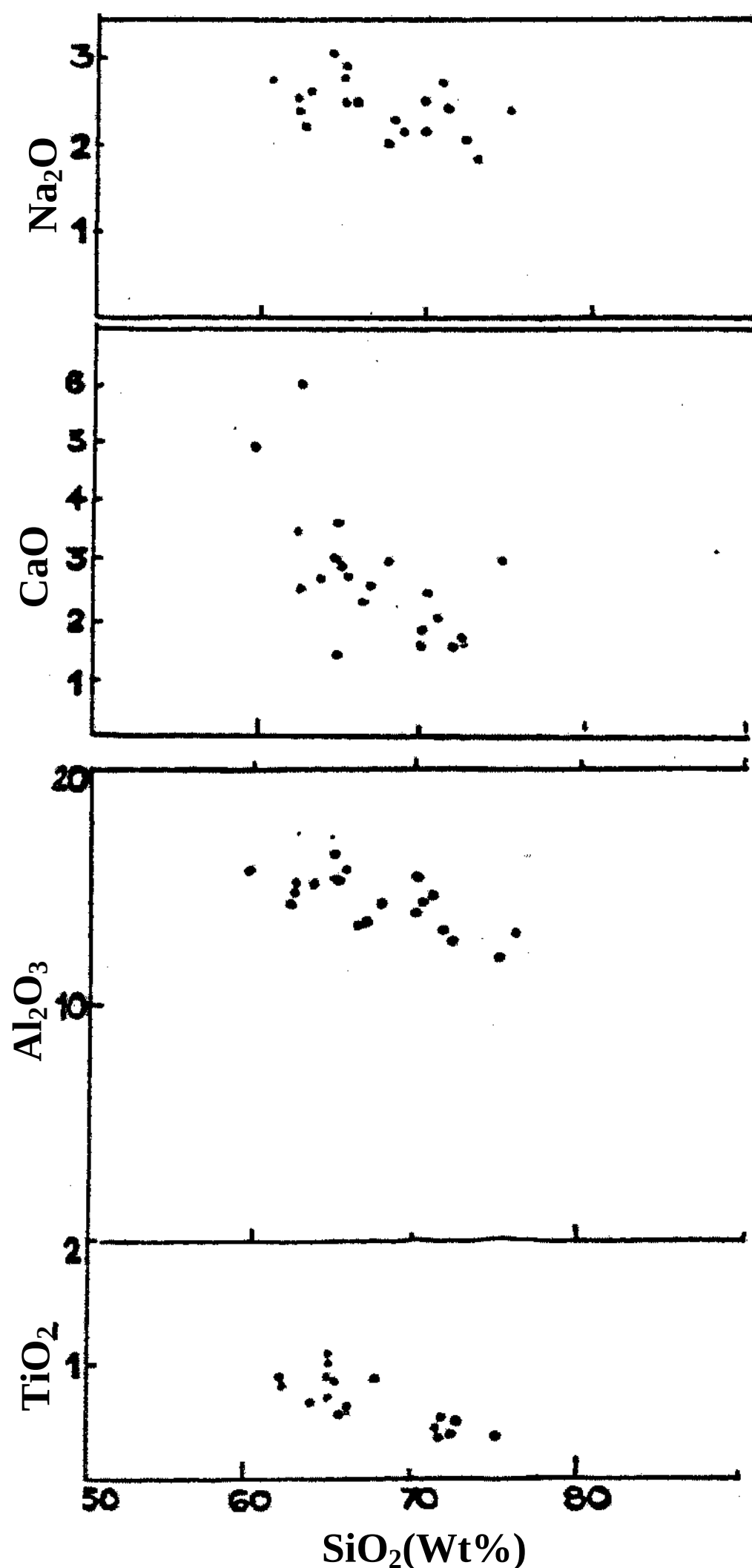


Fig 14: Harker variation diagram for SiO_2 vs selected oxides, all showing negative correlation with respect to silica in orthogneiss.

When plotted in the normative An-Ab-Or diagram (Fig 15), majority of the plots fall in the field of quartz-monzonite, adamellite and a few in granodiorite. In the normative Q-A-P diagram (Fig 16) in which A and P components were derived following the scheme of Le Maitre (1976), they plot in the granite field. On AFM diagram (after Irvine and Baragar, 1971) the felsic orthogneiss shows calc-alkaline trend (Fig 17). ACF diagram for the orthogneisses (after

Table I: Chemical Analyses of Orthogneiss																			
Sample No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
SiO ₂	71.80	64.99	72.50	75.05	68.01	70.06	64.85	59.68	71.00	63.73	66.59	62.59	62.35	67.01	62.27	70.12	65.22	64.88	70.50
TiO ₂	0.41	0.70	0.46	0.36	0.90	0.42	0.98	1.55	0.48	0.68	0.55	0.76	0.83	0.62	0.83	0.39	0.86	1.04	0.39
Al ₂ O ₃	13.07	16.53	12.61	12.00	14.42	15.49	15.78	15.79	14.59	15.17	13.42	14.74	15.82	13.57	14.36	13.93	15.36	15.48	14.32
Fe ₂ O ₃	3.74	4.07	3.52	3.73	5.73	2.24	4.75	6.22	2.37	4.60	6.97	9.41	4.06	6.33	8.95	3.60	6.73	5.95	2.83
MnO	0.04	0.05	0.09	0.06	0.06	0.03	0.05	0.07	0.03	0.06	0.09	0.13	0.05	0.07	0.12	0.05	0.09	0.04	0.04
CaO	1.50	2.94	1.64	2.98	2.97	1.77	3.57	4.92	1.98	2.63	2.30	2.54	2.63	2.51	3.44	1.53	2.87	1.37	2.40
MgO	0.23	1.20	0.41	1.15	1.76	0.68	1.37	0.97	0.47	1.18	0.15	0.16	1.23	0.32	0.45	0.40	1.52	2.07	0.69
Na ₂ O	2.07	2.78	1.67	2.38	2.22	2.24	2.48	2.67	2.44	2.57	2.06	2.37	2.55	2.30	2.53	2.54	2.92	3.14	2.72
K ₂ O	6.30	4.85	5.81	0.96	2.14	6.65	4.75	3.47	6.18	6.42	6.69	6.51	5.22	5.63	5.32	6.02	3.26	4.12	4.15
+H ₂ O	0.36	0.38	0.26	0.83	0.69	0.45	0.42	2.14	0.39	1.37	0.25	0.30	0.39	0.27	0.35	0.53	0.36	0.62	0.55
P ₂ O ₅	0.05	0.36	0.06	0.05	0.10	0.11	0.31	0.52	0.10	0.26	0.07	0.10	0.26	0.13	0.24	0.12	0.22	0.10	0.09
Total	99.57	98.85	99.03	99.55	99.00	100.14	99.31	99.00	100.03	98.67	99.14	99.37	98.72	98.76	98.86	99.23	99.41	98.81	98.61

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(Contd.)

Table I - *Contd.*

Sample No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
CIPW Norms																			
AN	28.89	34.21	35.40	41.79	42.84	29.84	42.77	48.00	30.75	33.38	30.40	34.05	34.47	34.11	36.14	24.05	34.13	18.78	32.9
Q	31.10	22.13	35.54	48.39	37.22	26.53	22.33	18.76	27.93	16.37	22.17	14.52	23.18	24.96	16.97	27.66	26.13	24.24	32.0
or	37.23	28.66	34.34	5.67	12.65	39.30	28.07	20.51	36.52	37.94	39.54	38.47	30.85	33.27	31.44	35.58	19.27	24.35	24.50
ab	17.52	23.52	14.13	20.14	18.79	18.95	20.99	22.59	20.65	21.75	17.43	20.05	21.58	19.46	21.41	21.49	24.71	26.57	23.02
an	7.12	12.23	7.74	14.46	14.08	8.06	15.69	20.85	9.17	10.90	7.61	10.36	11.35	10.08	12.12	6.81	12.80	6.14	11.30
C	0.24	2.22	0.74	1.75	3.29	1.65	0.81	0.00	0.53	0.00	0.00	0.00	1.82	0.00	0.00	0.74	2.34	3.60	1.20
di	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.12	0.00	0.37	2.95	1.41	0.00	1.33	2.96	0.00	0.00	0.00	0.00
hy	3.06	5.29	3.34	5.47	6.53	2.91	5.82	5.14	2.39	5.55	4.52	8.59	5.14	4.42	7.45	3.42	7.81	7.05	3.40
mt	1.81	1.97	1.70	1.80	3.48	1.08	2.29	4.42	1.14	2.22	2.97	3.28	1.96	3.07	3.38	1.74	3.42	3.68	1.30
il	0.78	1.33	0.87	0.68	1.71	0.80	1.86	2.94	0.91	1.29	1.04	1.44	1.58	1.18	1.58	0.74	1.63	1.98	0.70
ap	0.12	0.83	0.14	0.12	0.23	0.25	0.72	1.20	0.23	0.60	0.16	0.23	0.60	0.30	0.56	0.28	0.51	0.23	0.20

Geology of the payer-Weyprecht Mountains

Table II: Chemical Analyses of Amphibolite

Sample	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
SiO ₂	48.32	62.66	49.03	56.08	58.98	47.38	52.02	54.73	51.73	52.32	47.69	48.05	51.22	56.94	55.23	59.37	59.68	65.37	50.42
TiO ₂	3.04	0.79	2.63	0.91	1.60	2.63	2.34	2.82	2.73	2.80	1.28	2.67	2.92	1.66	2.62	2.37	1.18	0.89	0.91
Al ₂ O ₃	12.99	15.31	17.83	10.10	15.88	11.47	13.88	12.79	14.21	14.52	16.07	13.21	13.71	13.96	13.93	13.73	16.20	14.83	17.80
Fe ₂ O ₂	15.45	7.70	12.15	7.08	7.13	18.87	14.38	13.18	11.53	12.69	13.80	18.87	14.40	8.93	12.06	9.69	7.75	5.95	10.73
MnO	0.20	0.12	0.14	0.15	0.11	0.24	0.19	0.18	0.17	0.18	0.24	0.19	0.20	0.15	0.15	0.12	0.12	0.10	0.17
MgO	4.01	3.32	3.48	8.50	2.57	3.96	2.83	2.58	3.88	3.99	6.69	3.22	3.34	3.26	2.60	1.97	3.40	1.26	3.50
CaO	7.99	4.70	5.85	7.70	5.23	8.28	5.88	6.48	7.52	7.59	8.52	7.88	7.55	6.58	5.86	4.65	3.73	3.55	9.95
Na ₂ O	2.39	2.35	2.85	1.14	3.64	2.28	2.71	2.88	2.48	2.28	1.08	1.42	2.93	4.02	2.88	3.02	3.64	3.41	2.78
K ₂ O	2.18	2.20	3.42	5.10	2.44	2.58	2.99	2.68	2.67	1.68	2.63	2.68	2.45	2.35	2.24	2.58	2.88	2.81	1.00
P ₂ O ₅	0.85	0.11	0.94	0.56	0.49	0.75	0.64	0.82	1.33	1.25	0.13	0.47	0.74	0.62	0.73	0.65	0.37	0.24	0.22
H ₂ O ⁺	0.97	0.68	0.41	0.39	0.93	0.89	0.75	0.81	1.29	0.63	0.79	0.40	0.34	0.52	0.85	0.59	0.49	0.53	1.02
Total	98.39	99.94	98.73	97.71	99.00	99.33	98.61	99.95	99.54	99.93	98.92	99.06	99.26	98.99	99.15	98.74	99.44	98.94	98.50

(Contd.)

Table II — Contd.

Sample	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
CIPW Norms																			
AN	47.48	53.19	48.69	43.34	39.11	41.07	42.40	36.56	48.49	55.87	77.36	64.42	38.54	27.81	43.11	38.93	34.31	35.73	58.48
Q	4.40	23.14	1.67	6.16	14.69	0.00	6.66	11.98	8.86	12.29	0.00	4.61	5.60	8.66	14.21	20.17	13.45	24.78	2.88
or	12.88	13.00	20.21	30.14	14.42	15.25	17.67	15.84	15.78	9.93	15.54	15.84	14.48	13.89	13.24	15.25	17.02	16.61	5.91
ab	20.22	19.89	24.12	9.65	30.80	19.29	22.93	24.37	20.99	19.29	9.14	12.02	24.79	34.02	24.37	25.55	30.80	28.85	23.52
an	18.28	22.60	22.88	7.38	19.79	13.44	16.88	14.06	19.76	24.42	31.24	21.76	15.55	13.11	18.47	16.29	16.09	16.04	33.14
C	0.00	0.78	1.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.20	0.30	0.00
di	13.06	0.00	0.00	21.57	2.36	19.16	6.81	10.55	7.16	4.12	8.38	12.10	14.13	12.57	4.81	2.06	0.00	0.00	12.32
hy	13.16	14.24	14.43	15.44	6.94	16.68	14.36	7.94	10.58	13.95	23.74	18.67	9.71	6.47	9.78	6.64	12.91	5.76	13.15
ol	0.00	0.00	0.00	0.00	0.00	0.42	0.00	0.00	0.00	0.00	2.23	0.00	0.00	0.00	0.00	5.61	0.00	0.00	0.00
mt	6.58	3.32	5.99	3.49	4.49	5.99	5.57	6.26	6.13	6.23	4.03	6.05	6.41	4.58	5.97	4.50	3.89	3.47	3.49
il	5.77	1.50	5.00	1.73	3.04	5.00	4.44	5.36	5.18	5.32	2.43	5.07	5.55	3.15	4.98	1.51	2.24	1.69	1.73
ap	1.97	0.25	2.18	1.30	1.14	1.74	1.48	1.90	3.08	2.90	0.30	1.09	1.71	1.44	1.69	8.72	0.86	0.56	0.51

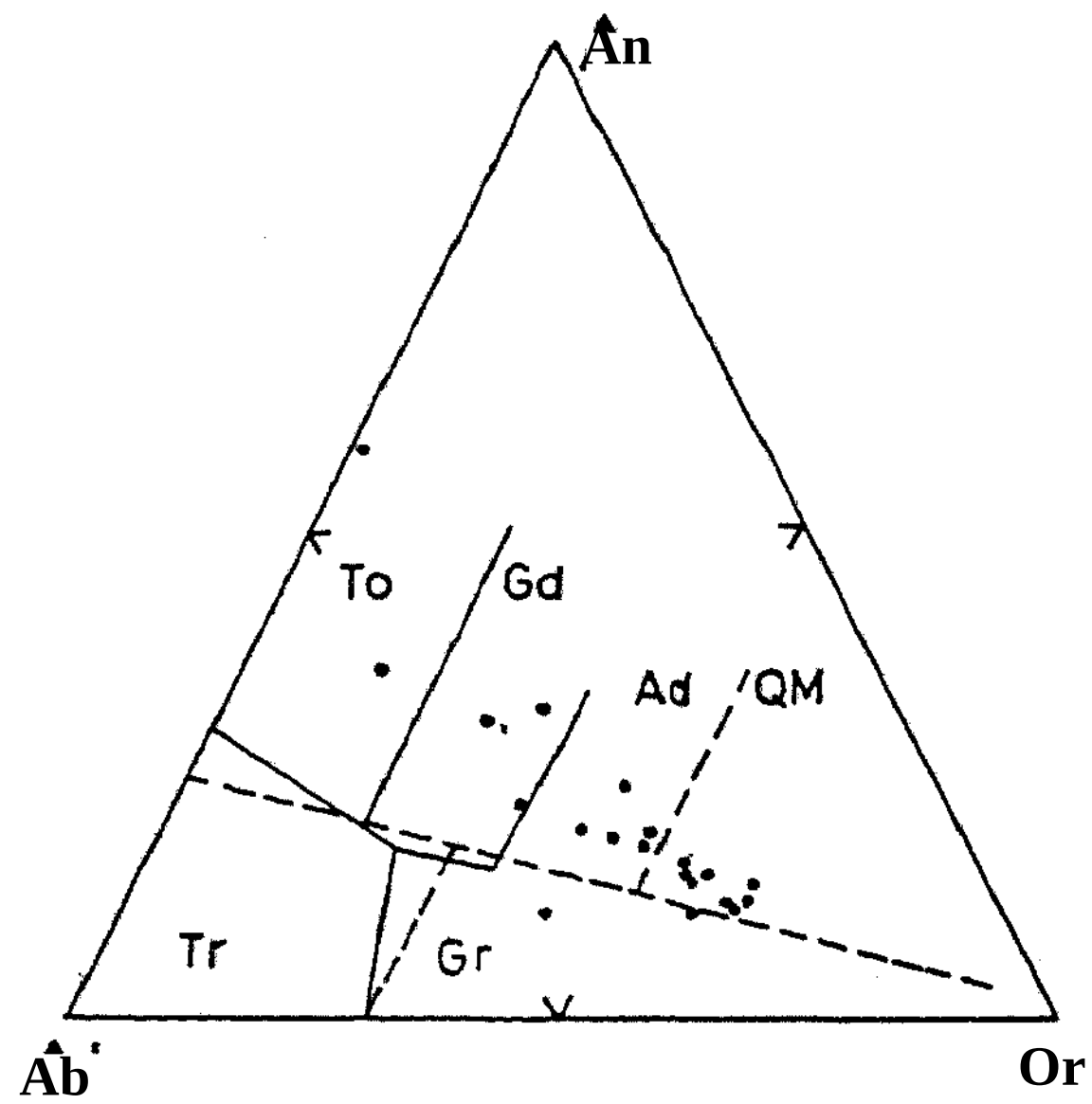


Fig 15 : Normative An- Ab-Or diagram for orthogneiss (after O'Connor, 1965 modified by Barker, 1979).

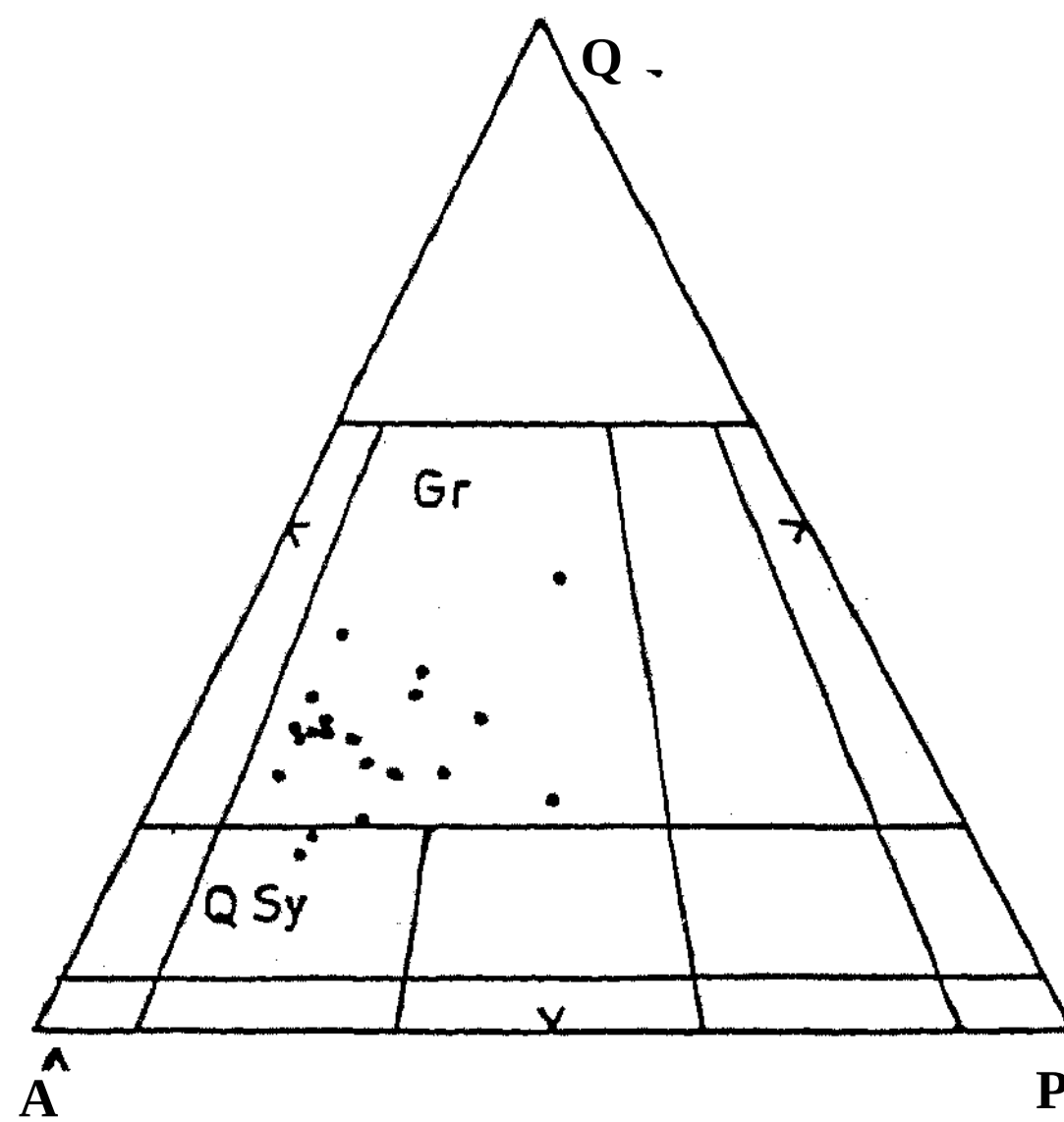


Fig 16 : Normative Q-A-P classification diagram for orthogneiss (after Le Maitre, 1976).

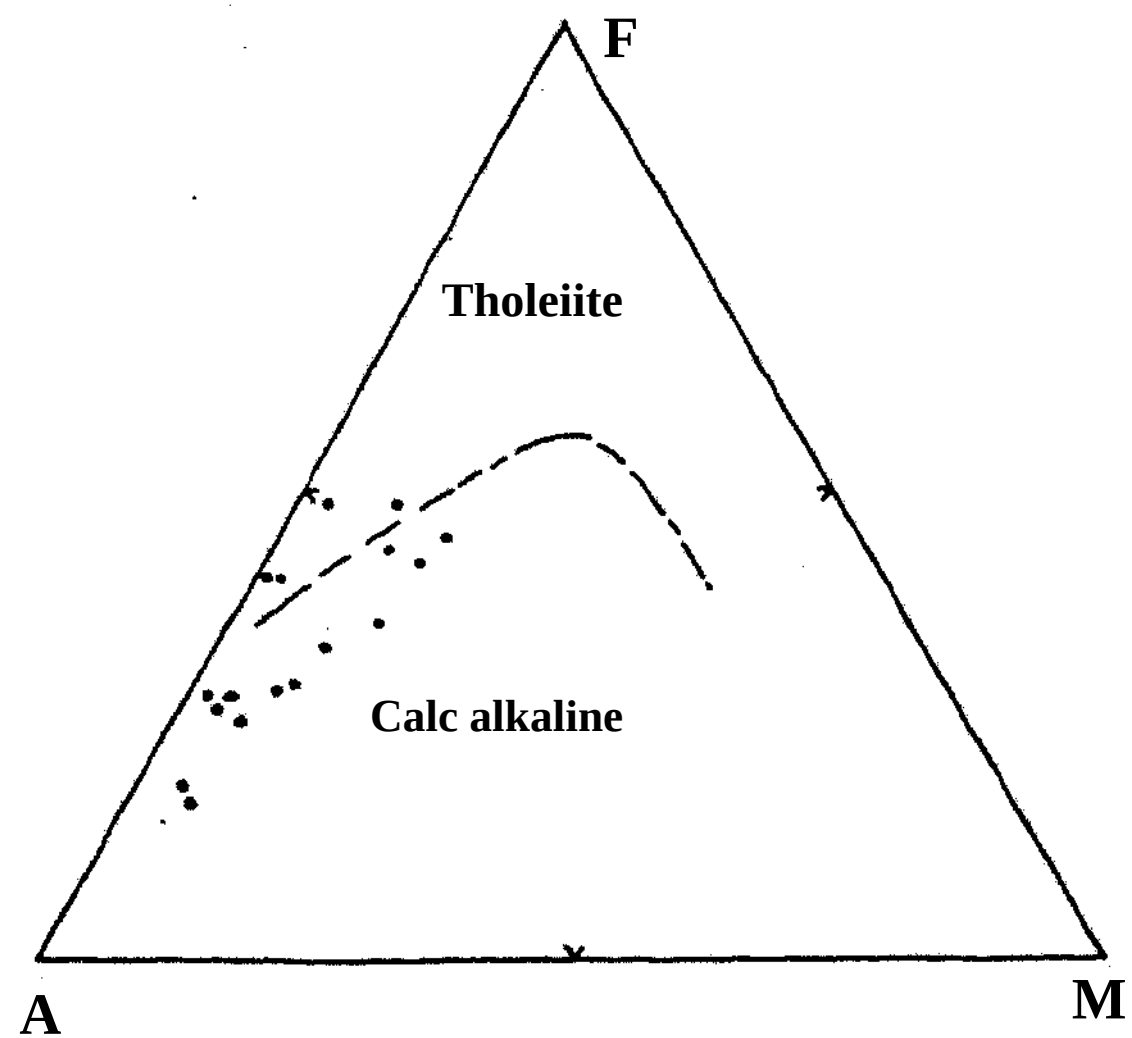


Fig 17: A FM diagram (after Irvine and Baragar, 1971) for felsic orthogneiss showing calc-alkaline trend.

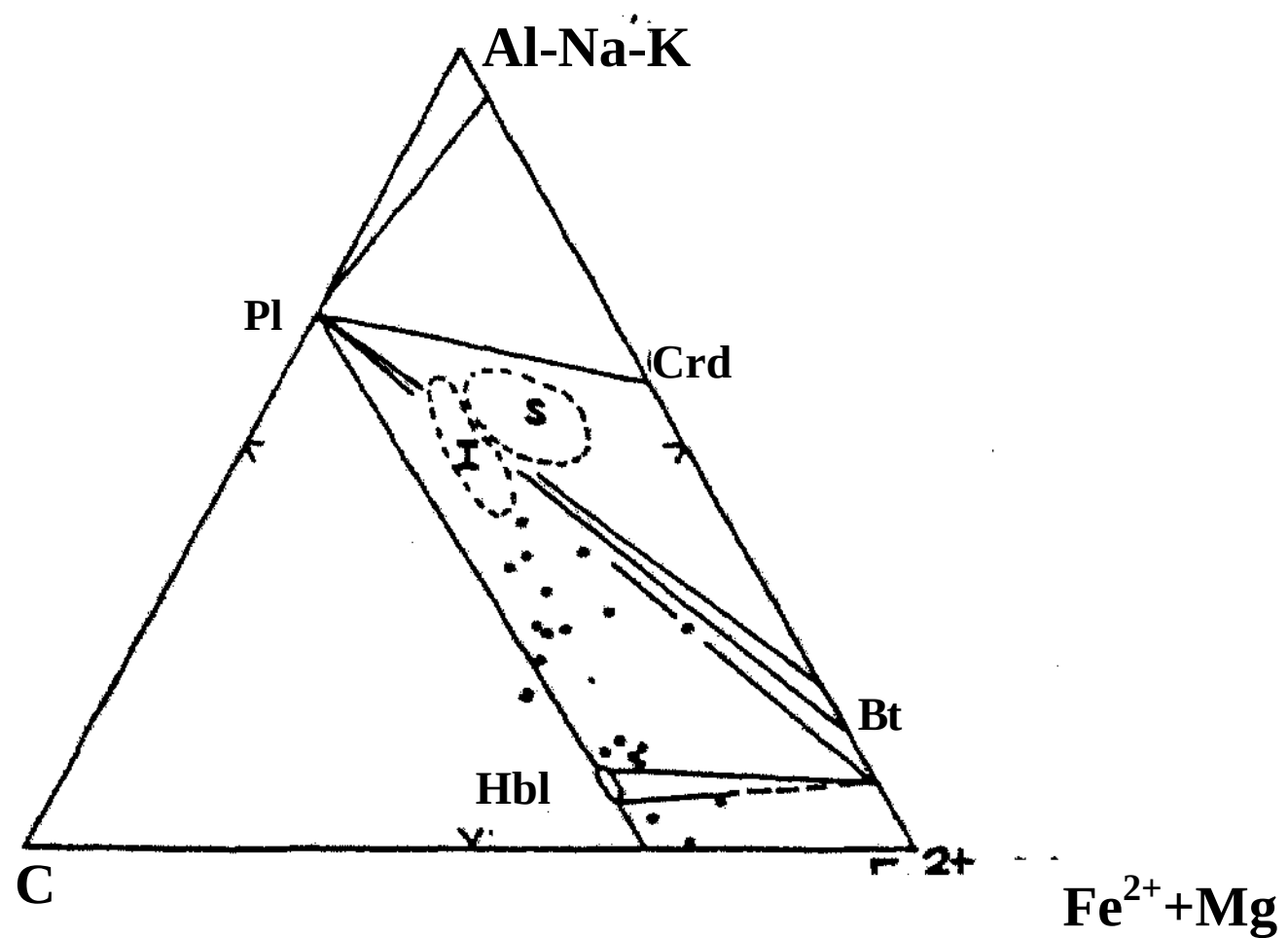


Fig 18: ACF diagram of orthogneiss (after White and Chappell, 1977) showing fields of selected I and S type granites.

White and Chappell, 1977) shows affinity towards hornblende-bearing I-type granites (Fig 18).

Considerable work has been done to assign the nature of source material particularly for granitoid or granitic gneisses by using the range of geochemical parameters, mineralogy and field associations (Chappell and White, 1974, White and Chappell, 1977). Application of these criteria for the orthogneiss of present area have shown dominance of I-type characters. Total absence of any association with metasedimentary lithology, and the presence of abundant hornblende and sphene in the mode strongly favour igneous parentage for the metagranite. The I-type nature is further strengthened by presence of meta-igneous (mafic) enclaves within these orthogneisses (Pitcher, 1982). However, analytical results when plotted on various source classification diagrams show combined nature of I and S types. Presence of normative diopside and corundum indicate metaluminous and weakly peraluminous nature of the orthogneisses, respectively. Similar nature is reflected in the diagram of Alumina Saturation Index vs SiO_2 (Fig 19). Discrimination factor of Shaw (1972) has positive value (3.09 to 0.07) for all the analytical results pointing towards igneous source.

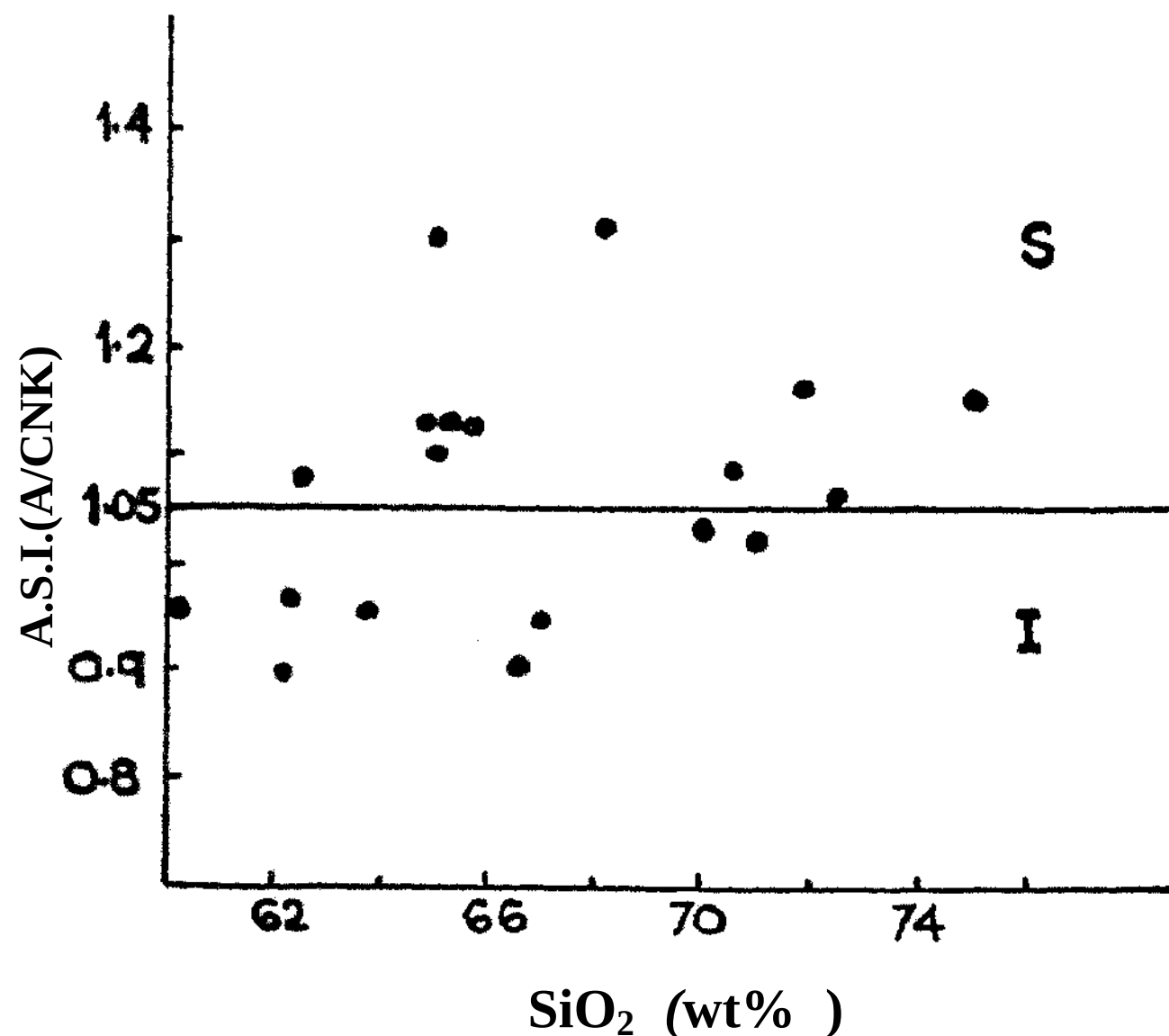


Fig 19 : Alumina Saturation Index{ $\text{ASI} = \text{molar Al}_2\text{O}_3 / (\text{CaO} + \text{Na}_2\text{O} + \text{K}_2\text{O})$ } vs SiO_2 in orthogneiss showing metaluminous to weakly peraluminous nature. Few are strongly peraluminous (discrimination after White and Chappell, 1983).

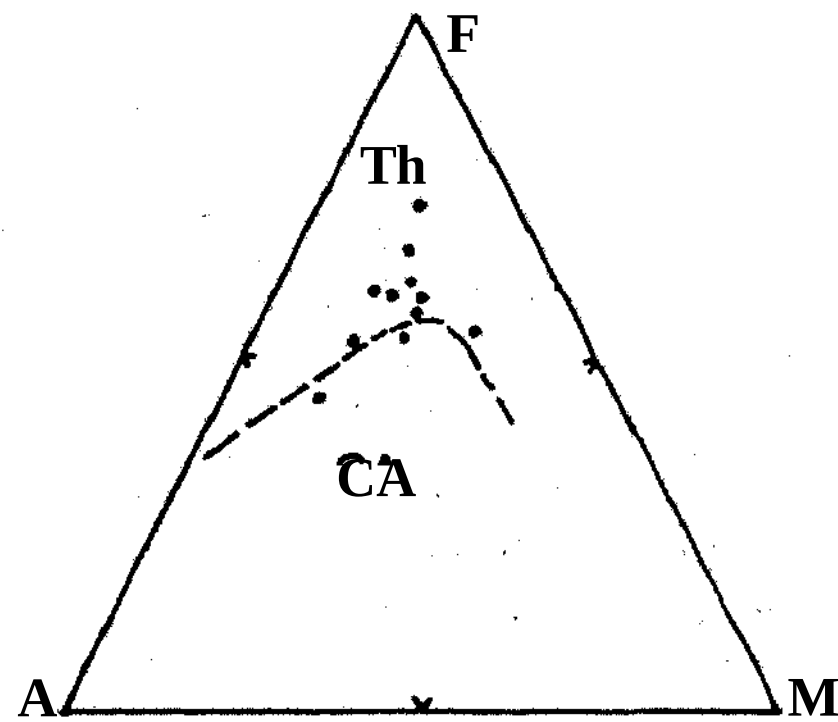


Fig 20: AFM diagram (after Irvine and Baragar, 1971) showing tholeiitic trends in mafic enclaves.

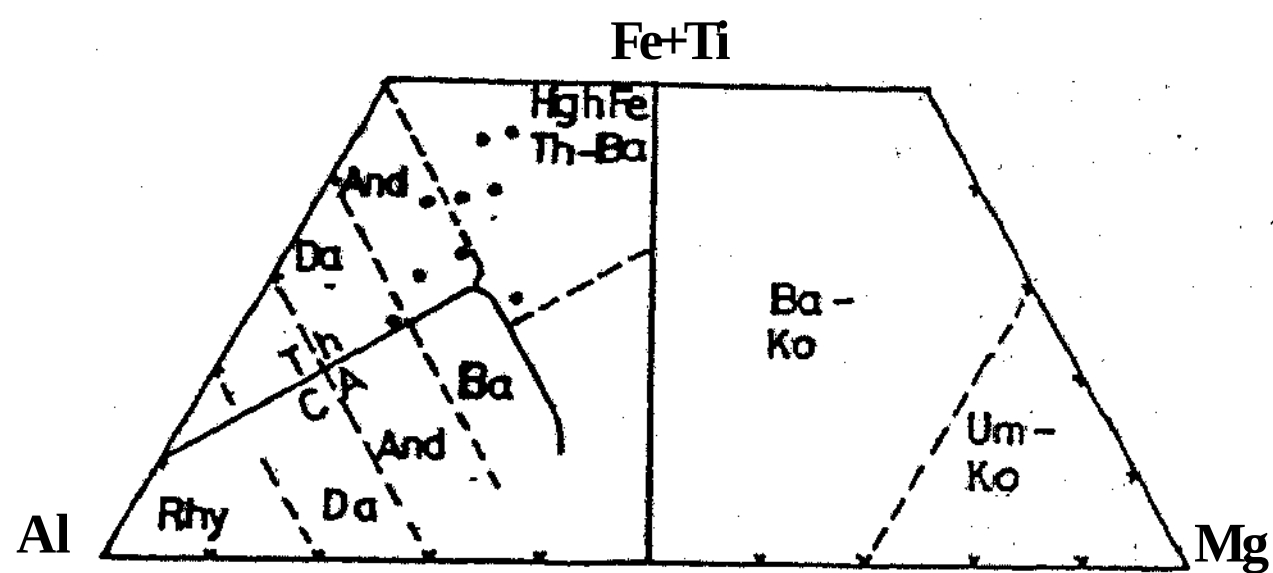


Fig 21: AFM cationic plots (after Jensen, 1976) showing tholeiitic nature of mafic enclaves.

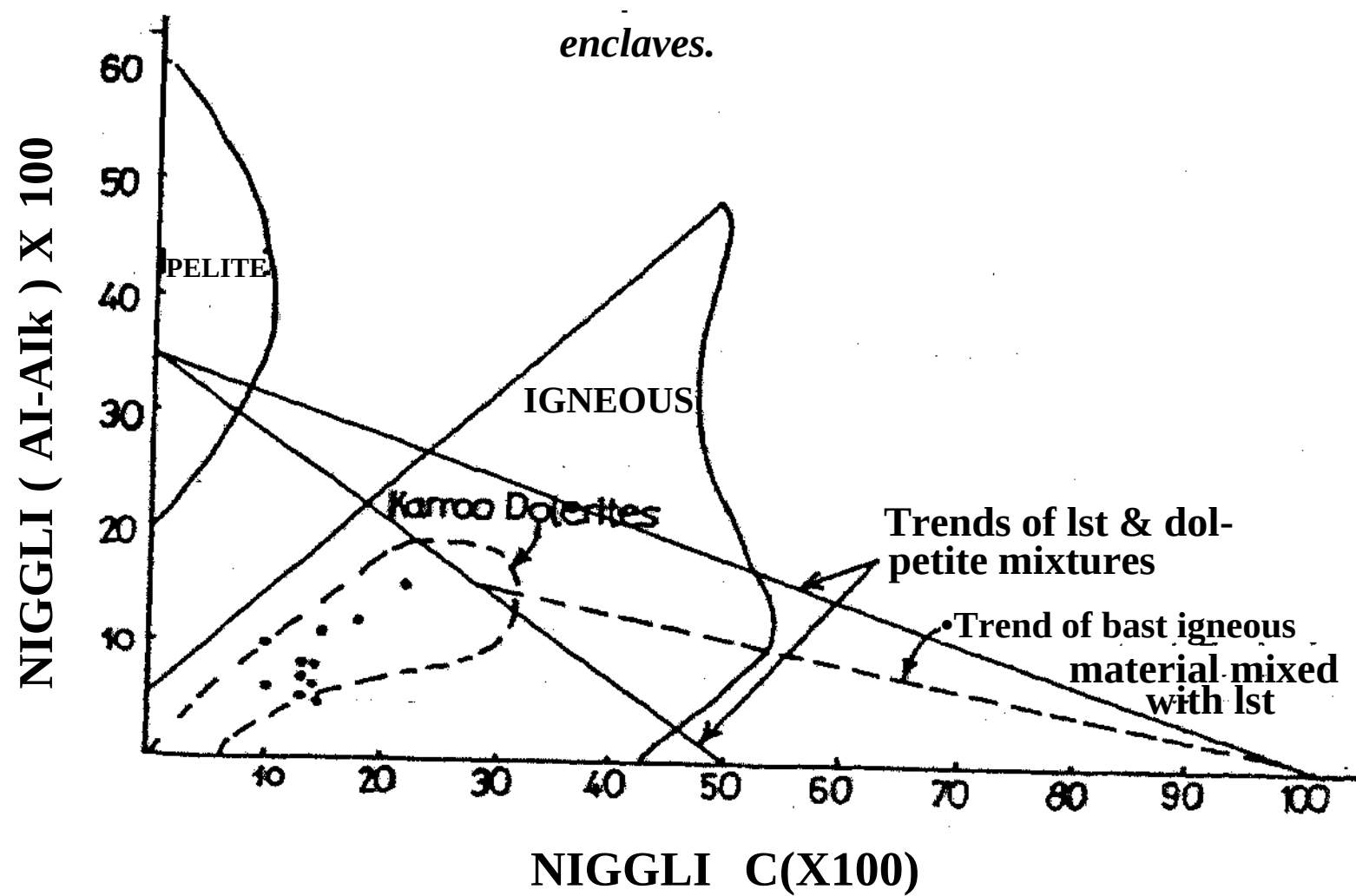


Fig 22 : -Plots of Niggli-C vs (al-alk) for mafic enclaves showing their meta-igneous nature (after Leake, 1964).

Pyroxene amphibolite

Nineteen rock samples were analysed and the results are presented in Table II. Comparison of the major oxides with the average of 200 amphibolites given by Poldervaart (1955) reveals that the pyroxene amphibolite has variable SiO₂ content, low MgO, CaO, Na₂O and high TiO₂ and K₂O. The major oxide chemistry are comparable with tholeiitic basaltic rocks. They also show tholeiitic affinity on AFM plots of Irvine and Baragar (1971) (Fig 20), and cationic plots of Jensen (1976) (Fig 21). In Niggli (al-alk) vs C plots all the pyroxene-amphibolite samples fall in the field of igneous rocks (Fig 22, after Leake, 1964).

Conclusion

Rocks of Payer-Weyprecht mountains reveal the presence of hornblende-biotite bearing orthogneiss with layered pyroxene- amphibolite enclaves. These have been profusely injected by pegmatite/aplite/granite veins. The mafic enclaves have preserved imprints of an earlier tectonothermal event at upper amphibolite- granulite facies. The orthogneiss show intrusive field relations with respect to mafic enclaves and were emplaced broadly syntectonic to D2 deformation event at amphibolite facies. Geochemically the gneisses are I-type granitoids showing calc- alkaline affinity.

Acknowledgement

We are indebted to Wg Cdr Dr. A.D.D'Souza for his excellent company and support in PayerCamp and to the fliers of Indian Navy who gladly ventured to take us into very difficult terrains for field work.

Shri M.K. Kaul, Director, GSI provided necessary facilities at Antarctica Division to carry out the work. Very useful suggestions of Dr. Abhinaba Roy of GSI helped us to improve the manuscript. We are grateful to them.

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